Long division

1 Simplify the following:

a $\dfrac{z+10}{10}$

b $\dfrac{9x^3+5x^5}{x}$

c $\dfrac{18x^3-12x^2+24}{3}$

d $\dfrac{48x^4+12x^3-30x^2-42x}{6x}$

2 Complete the following statement:

$$\frac{\square x^7-\square x^5}{2x^{\square}}=3x^4-4x^2$$

3 What polynomial, when divided by $2x^2$, produces $8x^6-5x^4+6x^2$ as a quotient?

4 Consider the division $\dfrac{x^3+1}{x+1}$.
What is the best way to write the polynomial x^3+1 in order to keep all of the like terms aligned during the long division process?

5 Divide x^2-6x+1 by $x+3$ using long division.

a State the remainder of this division.

b State the quotient of this division.

6 Perform the following using long division. State the quotient and remainder.

a $\left(x^2-8x+10\right)\div(x+4)$

b $\left(x^2-5x+11\right)\div(x+5)$

c $\left(x^2+3x+2\right)\div(x-3)$

d $\left(2x^3+7x^2+10x+8\right)\div(x+2)$

e $\left(3x^3-15x^2+2x-10\right)\div(x-5)$

f $\left(3x^3+2x^2+9x+6\right)\div\left(x^2+3\right)$

g $\left(x^4-81\right)\div(x+3)$

h $\left(4x^3-41x+15\right)\div(x-3)$

Synthetic division

7 Consider the division $\left(2x^2+3x-4\right)\div(x+4)$.
Is it possible to use synthetic division to perform this division?

8 Write each division in terms of the quotient and remainder by using synthetic division:

a $\dfrac{x^3+2x^2-29x-30}{x+6}$

b $\dfrac{x^3+x^2-12}{x-2}$

c $\dfrac{x^3-23x-10}{x-5}$

d $\dfrac{3x^3-19x^2+25x-22}{x-5}$

e $\dfrac{-3x^3+52x+17}{x+4}$

f $\dfrac{-3x^3-12x^2+28}{x+3}$

g $\dfrac{x^4-2x^3-11x^2-18x-9}{x-5}$

h $\dfrac{x^4-7x^3+7x^2+15x}{x-3}$

i $\dfrac{2x^4-3x^3-11x^2+6x}{x-3}$

j $\dfrac{x^4+9x^3+17x^2-13x+10}{x+5}$

k $\dfrac{x^5+32}{x+2}$

l $\dfrac{x^7-128}{x-2}$

9 Consider $x^3+6x^2+12x+16$.

a Use synthetic division to divide $x^3+6x^2+12x+16$ by $x+4$.

b State the polynomial $x^3+6x^2+12x+16$ in terms of the factor $x+4$ and the quotient.

10 Consider the statement $\dfrac{x^3-x^2-10x+6}{x+2}=ax^2+bx+c+\dfrac{d}{x+2}$.

a Use synthetic division to rewrite the division in terms of the quotient and remainder.

b Find $a+b+c+d$.

11 For each of the following polynomial functions:

i What does $P(5)$ represent?

ii Use synthetic division to divide $P(x)$ by $x-5$.

iii Now, determine $P(5)$.

a $P(x)=x^2-x-20$

b $P(x)=3x^3-18x^2+19x-16$

12 Consider $P(x)=x^3-x+2$.

a What does $P(0.5)$ represent?

b Use synthetic division to divide $P(x)$ by $x-0.5$.

c Now, find $P(0.5)$.

13 Consider $P(x) = x^4 - 6x^2 + 11$.

a What does $P\left(\sqrt{2}\right)$ represent?

b Use synthetic division to divide $P(x)$ by $x - \sqrt{2}$.

c Now, find $P\left(\sqrt{2}\right)$.

14 Consider $P(x) = x^2 + 2x - 8$.

a Use synthetic division to divide $P(x)$ by $x + 4$.

b Now, find $P(-4)$.

15 Consider $P(x) = x^2 - 9x + 21$.

a Use synthetic division to divide $P(x)$ by $x - 5$.

b Now, find $P(5)$.

16 Consider $P(x) = x^5 - 1$.

a Use synthetic division to divide $P(x)$ by $x - 1$.

b Now, find $P(1)$.

17 Consider $P(x) = x^3 + 4x + 2$. Use synthetic division to find $P\left(\sqrt[3]{3}\right)$.

18 Consider $P(x) = 3x^3 - 10x^2 - 12x + 16$.

a Use synthetic division to divide $P(x)$ by $x - 4$.

b Determine whether 4 is a zero of $P(x)$.

c Use synthetic division to divide $P(x)$ by $x - 8$.

d Determine whether 8 is a zero of $P(x)$.

19 Consider $P(x) = x^4 - 9x^3 + 22x^2 - 13x + 15$.

a Use synthetic division to divide $P(x)$ by $x - 5$.

b Determine whether 5 is a zero of $P(x)$.

c Use synthetic division to divide $P(x)$ by $x + 3$.

d Determine whether -3 is a zero of $P(x)$.

20 Consider $P(x) = x^2 - 7x + 12$.

a Use synthetic division to divide $P(x)$ by $x - 3$.

b Determine whether 3 is a zero of $P(x)$.

21 Consider $P(x) = x^2 + 3x - 11$.

a Use synthetic division to divide $P(x)$ by $x + 5$.

b Determine whether -5 is a zero of $P(x)$.

22 Consider $P(x) = 3x^3 + 4x^2 - 12x + 10$.

a Use synthetic division to divide $P(x)$ by $x + 3$.

b Determine whether -3 is a zero of $P(x)$.

23 Consider $P(x) = 3x^3 - 3x^2 - 6x$.

a Use synthetic division to divide $P(x)$ by $x - 2$.

b Determine whether 2 is a zero of $P(x)$.

24 Consider $P(x) = 2x^3 - x^2 - x - 3$.

a Use synthetic division to divide $P(x)$ by $x - 1.5$.

b Determine whether 1.5 is a zero of $P(x)$.

25 Consider $P(x) = 8x^3 + 2x^2 + 3x - 2$.

a Use synthetic division to divide $P(x)$ by $x - 0.25$.

b Determine whether 0.25 is a zero of $P(x)$.

26 Consider $P(x) = 3x^6 - x^4 + 2x^2 - 24$.

a Use synthetic division to divide $P(x)$ by $x - \sqrt{2}$.

b Determine whether $\sqrt{2}$ is a zero of $P(x)$.

27 Consider $2x^3 - x^2 - 9x - 4 = 0$.

- **a** Write down all the possible rational solutions.
- **b** Determine whether $x = \frac{1}{2}$ is a solution of the equation.
- **c** Determine whether $x = -\frac{1}{2}$ is a solution of the equation.
- **d** Rewrite $2x^3 - x^2 - 9x - 4$ in the form $(Ax + B)(Cx^2 + Dx + E)$.
- **e** Now, find all the solutions of $2x^3 - x^2 - 9x - 4 = 0$.

28 Consider $f(x) = x^3 - 6x^2 + 11x - 6$.

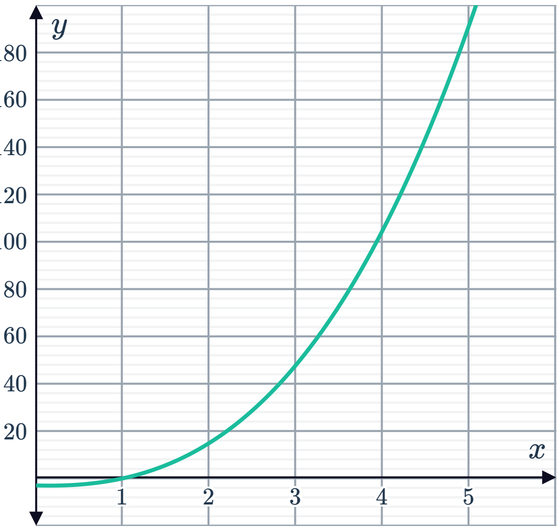
- **a** Write down all the possible rational zeros.
- **b** Determine whether $x = -1$ is a zero of $f(x)$.
- **c** Determine whether $x = 1$ is a zero of $f(x)$.
- **d** Factor $x^3 - 6x^2 + 11x - 6$ completely.
- **e** Now, find all the zeros of the polynomial $f(x)$.

29 Consider $P(x) = 6x^3 - 11x^2 + 6x - 1$ and the table of values for $P(x)$ on the integers in the interval $0 \leq x \leq 5$ as shown:

x	0	1	2	3	4	5
$P(x)$	-1	0	15	80	231	504

- **a** Use the table to find a solution to the equation $6x^3 - 11x^2 + 6x - 1 = 0$.
- **b** Use synthetic division to divide $P(x)$ by $x - 1$.
- **c** Determine whether 1 is a zero of $P(x)$.
- **d** Now, find all the solutions of the equation $6x^3 - 11x^2 + 6x - 1 = 0$.

30 Consider $P(x) = x^3 + 3x^2 - x - 3$ and the [redacted] of $y = P(x)$ shown:



a. Use the graph to find a solution to the equation $x^3 + 3x^2 - x - 3 = 0$.

b. Use synthetic division to divide $P(x)$ by $x - 1$.

c. Determine whether 1 is a root of $P(x)$.

d. Factor $x^3 + 3x^2 - x - 3$ completely.

e. Now, find all the solutions of the equation $x^3 + 3x^2 - x - 3 = 0$.

Applications

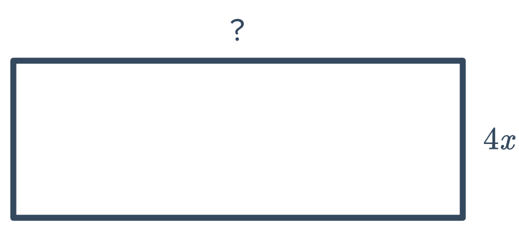
31 The rectangle has an area of $\left(0.2x^3 + 0.6x^2 + 0.45x + 0.1\right)$ units^2 and width of $(x + 0.5)$ units.

a. Find a rational expression for the length of the base.

b. Simplify the expression for the length of the rectangle.

32 The area of a parallelogram is given by $\left(x^4 - 7x^2 + 4x - 6\right)$ in^2, and its base measures $(x + 3)$ in in length. Find the perpendicular height of the parallelogram.

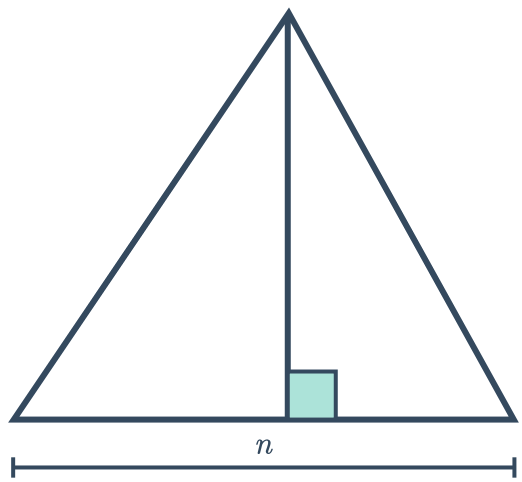
33 The rectangle shown has an area of $\left(4x^4 - 8x\right)$ units^2.
Find a polynomial expression for its length.



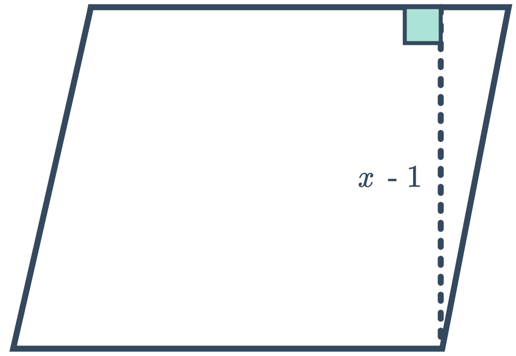
34 The rectangle shown has an area of $\left(4x^4 - 8x\right)$ units^2.
Find a polynomial expression for its length.

35 The rectangle shown has an area of $\left(5x^3 + 7x^2 - 18x - 8\right)$ units^2.
Find a polynomial expression for its length.

36 The triangle shown has an area of $\left(13n^3 + 11n^2 + 29n\right)$ units^2.
Find a polynomial expression for the perpendicular height.



37 This parallelogram has an area of $\left(2x^3 + 4x^2 - 5x - 1\right)$ units^2.
Find a polynomial expression for the length of its base.



38 If the distance traveled is $\left(2x^3-2x^2+5x+9\right)\ \mathrm{mi}$ and the speed is $(x+1)\ \mathrm{mi/h}$, find the time traveled in hours.

39 It costs $\left(4x^5+5x^4+2x^3+10x^2-5x+14\right)$ dollars to replace the lawn in the backyard. If the new lawn costs $(x+2)$ dollars per square meter, find the area of the lawn.

40 The amount of dust, in milligrams, collected on a table after x days without dusting is modeled by the function $f(x)=9x^3-30x^2-34x+60$. After how many days without dusting will the table have collected $20\ \mathrm{mg}$ of dust?